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Introduction

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What is this book about? It is not about how to utilise the hazard analysis critical control point (HACCP) system which has been covered many times in numerous publications from international, national and trade organisations and from individual practitioners on a world-wide basis (see section 1.1). Nor is it an account of the role of HACCP in any particular sector of the food industry. This book chronicles the implementation of HACCP in a broad range of situations and asks the following questions:

- What lessons can we learn?
- How could we have done things differently or better?
- How can we make the most of HACCP now and in the future?

1.1 The development of HACCP

The development of HACCP from its initial conception by Pillsbury working together with Natick is well documented (Bauman 1974, 1990; US Department of Health, Education and Welfare 1972). Initially developed as a means of obtaining increased confidence over the microbiological safety of foods used in the space programme, the potential of the technique for ensuring control of food safety hazards in a broad range of foods in manufacturing, distribution, food service and retail situations was soon realised. Although the HACCP concept continued to develop, it did so in a somewhat piecemeal fashion, and it was not until the mid- to late 1980s that HACCP development began to accelerate in a coordinated way. The HACCP concept was endorsed by WHO/FAO as an effective way of controlling foodborne disease in 1983 when the Joint FAO/

WHO Expert Committee on Food Safety advised that HACCP should replace the traditional end product testing approach to food safety assurance (FAO/WHO 1983). Since that time many other respected bodies have published guidelines on the use of HACCP (ICMSF 1988, US National Advisory Committee on Microbiological Criteria for Foods 1992, ILSI 1997, Campden & Chorleywood Food Research Association 1997).

It was only during the late 1980s and beginning of the 1990s that there was a concerted attempt to harmonise, on an international scale, the HACCP approach and the terminology used. Initially, interested groups in the USA, Canada, Europe, Australia and New Zealand pursued the development of HACCP most actively, particularly those representing the larger food manufacturing industries. Although the major thrust for the development of HACCP has always been the management of food safety (primarily by those responsible for food manufacturing), increasingly HACCP has been looked upon as a regulatory tool for the enforcement of food safety. Prior to the early 1990s HACCP developments concentrated upon the principles of HACCP rather than the practical details of HACCP application or implementation. The current Codex Alimentarius HACCP document (CAC 1997) is really the first truly authoritative, internationally agreed document that addresses not only the agreed HACCP principles but also guidelines for its application. This document clearly identifies that the HACCP system consists of the seven principles listed in Table 1.1.

The guidelines for the application of the HACCP system are also clearly outlined in 12 tasks, as shown in Table 1.2. Most people today regard this Codex document as the international benchmark for HACCP.

1.2 HACCP in practice

As stated above, the primary movers in the development of the HACCP system tended initially to be the larger food manufacturers and, latterly, those with a

Table 1.1 Seven principles of the HAACP system

Principle 1	Conduct a hazard analysis
Principle 2	Determine the critical control points (CCPs)
Principle 3	Establish critical limit(s)
Principle 4	Establish a system to monitor control of the CCP
Principle 5	Establish the corrective action to be taken when monitoring indicates that a particular CCP is not under control
Principle 6	Establish procedures for verification to confirm that the HACCP system is working effectively
Principle 7	Establish documentation concerning all procedures and records appropriate to these principles and their application

Table 1.2 Guidelines for the application of the HACCP system

Task 1	Assemble the HACCP team
Task 2	Describe product
Task 3	Identify intended use
Task 4	Construct flow diagram
Task 5	On-site verification of flow diagram
Task 6	List all potential hazards, conduct a hazard analysis, determine control measures
Task 7	Determine CCPs
Task 8	Establish critical limits for each CCP
Task 9	Establish a monitoring system for each CCP
Task 10	Establish corrective action for deviations that may occur
Task 11	Establish verification procedures
Task 12	Establish record keeping and documentation

regulatory perspective. Although as stated above, HACCP has now been applied to all sectors of the food supply chain, there has been no attempt to identify the lessons to be learned from such a broad-based application. This book seeks to address this issue by inviting contributions from a range of practitioners from all parts of the food industry.

Contributions have been made by authors practising in a wide range of HACCP-related roles across the supply chain including small, medium, and large sized food manufacturers, covering primary conversion and manufacturing, food retailers and those with involvement/responsibility for HACCP training and regulatory enforcement. Contributors were asked to structure their contribution broadly as shown in Table 1.3 and to consider a number of key questions when putting their chapter together.

The issues described in Table 1.3 are most relevant to those organisations implementing HACCP. Those for enforcement agencies/bodies and retailers dealing with manufacturers were considered likely to be different. The brief to these contributors is described in Table 1.4.

The authors have been drawn from a wide geographical basis and from enterprises within a wide range of cultural backgrounds and have been encouraged to outline their thoughts in their own personal style. The reader should therefore accept that the value of the contribution lies in the broad range of experience outlined in the content rather than in a uniform presentation style.

1.3 Key themes

Two contributors to the book address the issues of HACCP application in a primary conversion food sector and look to identify the lessons learned. While

Table 1.3 Key questions for contributors from organisations implementing HACCP

Introduction	• A brief introduction to the industry in which the organisation is involved and any distinctive hazards the industry faces • An outline of the model for HACCP implementation that was followed where appropriate
Planning for HACCP	• How did the HACCP approach fit with corporate strategy? Where was HACCP implementation in the priority list? • How did the business plan its HACCP implementation programme? (e.g. who was in the project team, what timetable/ milestones were established, what kind of budget was available?) • What information was available to the business (e.g. prerequisites, good manufacturing practices, scientific data, references, etc.) and did this prove sufficient? • In what ways did the original plan (the timetable, personnel involved, etc.) change and why?
Implementing HACCP	• What were the major challenges in implementation and how were these overcome? As an example, was hazard analysis the most complex stage? If so, how was it tackled? • What involvement, if any, did the business have with enforcement agencies/bodies in the development of the business's HACCP system? • Did the organisation anticipate all the issues raised by auditors/ enforcement agencies?
Operating the HACCP system	• What benchmarks were used to judge success and how has the system measured up to those benchmarks? • Were there any unexpected difficulties arising from HACCP implementation? How were they managed? • What benefits has the implementation of HACCP brought to the business? • What is the most demanding aspect of running a HACCP system? How is this managed?
Developing the HACCP system	• Has the way HACCP relates to corporate strategy changed since implementation? • What element in your HACCP system has needed development most in the light of experience? • How has the HACCP system changed/evolved since implementation? • How have changes been integrated into the system, e.g. new products or technology? • How has the HACCP system been integrated with other systems, e.g. ISO 9000? • In what ways has HACCP been extended into new product design? • How has the business kept the system up to date (e.g. new hazards, keeping staff motivated)? • Has the implementation of HACCP changed the relationship with the enforcement agencies/bodies? If so, why?
Conclusion	• What was the most unexpected (positive and negative) aspect about implementing and running a HACCP system? • What would you do differently, given the benefit of experience? • Comment on whether the implementation of HACCP has 'added value' in a product safety sense. • What is the future for HACCP?

Table 1.4 Key questions for contributors from enforcement agencies and retailers

Introduction	• Brief description of regulatory framework and nature of enforcement agency, or retail sector and manufacturer relationship. • Brief description of the nature of a HACCP inspection (e.g. stages followed, information expected) or requirement.
Assessment of the HACCP system	• What do you look for (e.g. prerequisites) to see how well the system is working in practice as well as on paper? • Do you use a check-list of key issues/questions to test the effectiveness of a HACCP system? If so, are you happy with this as an approach? • What makes it easier to carry out an effective HACCP inspection? What can organisations do to facilitate your work? • What are the most common areas of weakness in the HACCP systems you have inspected? Why do you believe these problems are widespread? • What are the most common areas of improvement you suggest?
Developing HACCP assessment	• How did your own skills and competencies, and those of your agency/organisation, match up to those required for HACCP system enforcement? • How did you overcome any deficiencies in skills and competencies? • How did you/do you determine your roles and responsibilities with respect to HACCP enforcement? Is there any conflict with industry? If so, how should this be resolved? • Has HACCP enforcement changed the nature of your relationship with industry? If so, in what way? • Do you believe that current inspection/assessment procedures are adequate to determine effective HACCP implementation? If not, how do you believe they should change? • How do you see your role in HACCP enforcement developing in the future (e.g. partnership with industry)?
Conclusion	• Has HACCP implementation affected the role of your agency/organisation with respect to product safety? If so, please elaborate and discuss how you see this developing in the future. • Do you believe that HACCP implementation offers real advances in product safety from the enforcement or retailer standpoint? • What do you believe are the real weaknesses of HACCP as a product safety system, viewed from the enforcement or retailer standpoint? How should these weaknesses be overcome? • What do you see as the future for HACCP viewed from the enforcement or retailer standpoint?

HACCP principles remain the same throughout, the issues to be tackled when considering application to a primary conversion sector of the food chain will be somewhat different from those issues to be tackled when applying HACCP to the manufacturing sector of the supply chain.

The applicability of the HACCP system to small and medium sized enterprises (SMEs) has always been questioned as the formalised, technically based full HACCP system outlined in the 1997 Codex document places severe demands on SMEs in terms of technical data and skilled resource for hazard analysis and time. Although there are a number of publications and studies tackling the HACCP and SMEs issue (WHO 1993a, 1999a; Department of Health 1993), this is an area of HACCP application that has still not been optimised and agreed. This book contains two contributions from practitioners addressing the application of HACCP in SMEs, one from an SME manufacturer and one looking at the topic from a more academic background. The effective application of HACCP principles in SMEs represents a significant opportunity to raise the standard of food hygiene in a very large sector of the supply chain. It is clear, therefore, that we need to take account of all the lessons learned so as to maximise the chance of success. The contributions from the larger organisations demonstrate that they too have not always got it right first time, and have learnt lessons that can be shared with others.

Another issue not tackled during HACCP development is that of the impact of different cultural and language backgrounds; thus successful application of HACCP relies on a cross cultural/linguistic consensus on the meaning and definition of these terms. Translation of key HACCP terms such as ‘hazard’, ‘risk’, ‘control measure’, etc., is fraught with potential for genuine misunderstanding. The success of the Codex approach depends to some extent on an open questioning and debating approach with a number of key individuals contributing in public debate. This can be somewhat adversarial in nature and is essentially a ‘Western style’ approach. Such an approach is, however, anathema to many countries, particularly those in Asia and India where open debate is frowned upon. The book contains two contributions (one from Thailand and one from India) where open debate is not the accepted norm and these chapters illustrate the mechanisms used to operate HACCP successfully in a ‘non-Western’ cultural and language background.

The history of the development of HACCP clearly demonstrates that the HACCP system is continually evolving. This is entirely to be expected as HACCP must evolve in response to new information (e.g. allergens as potential hazards), new food processing technologies (e.g. microwave pasteurisation) and the realisation that techniques must be available to assess the effectiveness of HACCP. In the latter case it is only recently that the different roles of verification and validation in determining the effectiveness of HACCP have become clear (ILSI 1999). In the same way the role of prerequisites or good manufacturing/hygiene practices (GMP/GHP) standards has developed over a period of time to the point where today prerequisites and/or GMP/GHP standards are accepted as the essential foundation upon which HACCP must be built, and without which HACCP implementation is unlikely to be successful.

Underpinning all the above is the need for training in both HACCP and supporting prerequisite good hygiene practice. The former is a requirement clearly identified by the WHO (1993b, 1996, 1999b) and FAO (1998). There are

a large number of publications in addition to the above dealing with HACCP training (Mortimore and Wallace 1998, Pierson and Corlett 1992, Engel 1998, de Winter 1998, Bryan 1991, Agriculture Canada 1995) and many on government/regulatory internet sites in themselves demonstrating how HACCP has evolved. It is relevant to note, however, that it is only recently that progress has been made in developing a HACCP Training Standard (Royal Institute of Public Health and Hygiene 1998). The book contains one chapter looking specifically at ways of maximising the benefit of HACCP training in a developing, continually evolving system.

At the present time within the USA the Food and Drug Administration (FDA) has mandated HACCP for fish and fishery products and is proposing mandating HACCP for fruit and vegetable juices. The United States Department of Agriculture (USDA) has already mandated HACCP/pathogen reduction requirements for the meat and poultry industry and the US National Marine Fisheries Service (NMFS) seafood inspection programme operates a voluntary HACCP programme for seafood plants. The USA has taken the lead in the use of HACCP as a regulatory tool and other countries look set to follow with a mandatory HACCP approach. There is a growing realisation among regulatory authorities that the use of HACCP by the food manufacturing industry provides them with both an opportunity and a challenge. HACCP implementation provides regulatory authorities with the opportunity for them to make meaningful assessments of product safety management systems by making use of HACCP records. The challenge for regulatory authorities is for them to gain the skills and competencies needed to make the proper assessment of HACCP. The recent FAO/WHO *Expert Consultation on the Role of Government Agencies in Assessing HACCP* (FAO/WHO 1998) addressed the above issues in some detail. The book editors believed at the outset that it was important that this book should contain contributions from individuals with regulatory responsibility for HACCP so that the regulatory point of view, the other side of the 'HACCP coin', could be documented. Accordingly this book contains contributions from experienced individuals well qualified to debate the issues surrounding the enforcement/assessment of HACCP.

One of the major developments of the last 20 years has been the increasing globalisation of trade. This has led to the availability of fresh and processed produce, of ever increasing variety, all year round. Underpinning this globalisation of trade has been the recognition that countries need to be allowed to trade with each other without unnecessary barriers to trade. The World Trade Organisation (WTO) sanitary and phytosanitary (SPS) agreement (WTO 1995a, 1995b), which came into force in 1995 and applies to all sanitary and phytosanitary measures that can or may affect international trade, has gone a long way towards empowering countries to trade freely with each other but increasingly countries are looking to develop and implement the concept of 'equivalence' to further enhance trade. The concept of equivalence recognises that although different countries can have different food safety systems, they can in fact offer equivalent levels of food safety protection. A number of countries

have already negotiated equivalency agreements with trading partners, thereby facilitating and simplifying international trade. One of the key considerations of international trade must be the safety of the food being traded. The globalisation of food trade and the recognition that 'emerging hazards' are a significant issue has increased the risk of cross-border contamination of food. The potential for significant transference of contaminated food between countries and/or regions is therefore much increased with potentially catastrophic consequences for public health. The standards, guidelines and recommendations adopted by Codex for food safety are recognised in the WTO SPS agreement as the basis for the harmonisation of sanitary measures and HACCP is a central food safety system of choice promulgated by Codex. It is inevitable therefore that HACCP will play an increasingly important food safety role, not only as the system of choice within individual countries, but also on the world-wide stage as an important contributor to the facilitation of world trade in foodstuffs. Although this book does not contain contributions that specifically address the increasingly important role of HACCP with respect to trade, the international list of HACCP contributors clearly demonstrates the global impact that HACCP is having, and the messages derived from this book, e.g. 'Making the most of HACCP' are therefore particularly relevant.

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